Task : 1

1. Was the IP address always the same?

Yes. This means your computer used the same Google server for those requests

2. Did different students get different IPs?

Different students may see different IP addresses because:

- Google has many servers around the world.
- The DNS server returns the IP address of a nearby or less busy Google server.
- Students may use different ISPs, Wi-Fi networks, or locations

3. Why do you think this happens?

Google uses **DNS** and **load balancing** to direct users to different servers.

Reasons include:

- Different geographic locations.
- Different Internet Service Providers (ISPs).
- Server load balancing.
- IPv4 vs IPv6 availability.

4. What is RTT (Round Trip Time)?

RTT (Round Trip Time) is the total time it takes for a data packet to:

1. Travel from your computer to the destination server.
2. Return from the server back to your computer.

It is measured in **milliseconds (ms)**.

Task 2

	Amazon	NSG Academy	Flipkart	Netflix
Domain Name	amazon.com	nsgacademy.com	flipkart.com	netflix.com
Name Server	Ns-1447.awdns52.org	Ns1249.websitewelcome.com	Sdns14.ultradns.bi2	Ns.1372awdsn.43org
Hosting Provider	Amazon technology	HostGator.com	Flipkart Internet	Amazon Technology
SSL	Amazon RSA	YR2	Global Sign RSA	Digicert
CDN	Hidden	Hidden	Hidden	Hidden
Technology use	Hidden	Hidden	Hidden	Hidden
IP Address	64:ff9b:6257:aa4d	64:ff9b:c6b9:9bb3	64:ff9b:67f3:205a	2a05:do18:76:b684:b34d

1) Who owns the domain

Ans)Amazon – Amazon technology

NSG- Now Solution Group

Netflix- Netflix

Flipkart-LLC

2) Where it is Hosted
   Ans)Amazon –Amazon Technologies

   NSG- HostGator.com

   Netflix- Amazon Technologies

   Flipkart-Flipkart Internet

Task 3

★Google: Registration Date: 1997-9-15
Expires on 14-9-28
Registar: Mark Monitor
DNS record: A : 142.250.72.14
AAA: 2607 :F860:4006:800:: 2000

☆TCS: Register On: 25-11-1997
Expires on 24-11-20341
Registrar: Network Solutions.
DNS recerd: 4.247.162.6.

*Infosys: Registor on : 17-74-1992
Expire on 13-10-2026
Registror: CSC corporate Domains
DNS record: 35.71.138.178

Microsoft: Register on : 2-5-1991
Expire on 3-5-2027
Registrar: Mark Monitor
DNS record: A: 150.171.104.22

NSG: Rezistor on : 14-1-2015
Expire on 14-1-2027
Registrar: Rebel Ltd
DNS Record: 192.185.155**.129**

QI)why does one company own multiple domains :
1)To protect their brand nom
2) To reclirect user to the main websites
az what

Q2)happens if a domain expires?
1) website may stop working.
2) someone else may register the domain

Q3) why don't companies use free domains?
1) They look less professional
2) Paid domains give companies better control

Task 4

College Project → Netlify or Vercel (Hobby/Free tier)

Zero cost, zero DevOps, and something that just works the moment you push to GitHub. Free tiers give instant deploys, free SSL, and a live URL in minutes — perfect since college projects are non-commercial by nature and rarely need a real production database.

Startup MVP → Railway

An early MVP needs a small team, a backend + database, and unpredictable/low traffic while you look for product-market fit. Railway's native multi-database support and true usage-based (scale-to-zero) pricing keeps costs low pre-revenue while still feeling production-grade once you get real users.

E-commerce Website → Render (or Vercel for a headless/Next.js storefront)

E-commerce needs reliable always-on uptime (no cold starts costing sales), predictable monthly costs, a real database, and solid multi-region performance. Render's fixed-ish plans and native Postgres fit well; Vercel is strong for a headless Next.js frontend layer.

Banking Application → AWS EC2

A banking app has strict requirements around compliance (PCI-DSS, data residency, auditing), custom network isolation, and fine-grained security control. Only AWS EC2, paired with the wider AWS ecosystem (RDS, IAM, VPC, KMS, CloudTrail), gives the level of infrastructure control and compliance tooling that regulators typically demand.

Task 5

1. What is the difference between what you SEE on screen and the raw code?

A) The webpage shows the final visual content such as menus, buttons, images, and text. The raw source code contains the HTML instructions used by the browser to create and display that webpage.

2. Why do developers leave comments in their code?

A) Developers leave comments to explain parts of the code, making it easier to understand, maintain, and modify later. Comments are not normally displayed on the webpage.

3. What do you think the <title> tag is used for?

A) The <title> tag defines the title of the webpage shown in the browser tab. In this source code, the title is **“State Bank of India.”**

Task 6

1. Did your changes disappear after refreshing? Why?
Yes, they did — and that's expected. When you use "Inspect" (DevTools) to edit a page, you're only editing the **copy of the page loaded in your browser's memory** — not the actual file sitting on the server.

2. Did the website change for anyone else in the world? Why not?
No. Your edits only existed inside *your* browser, on *your* computer, for that one browsing session. Nothing you did with Inspect ever got sent back to the server.

3. What's the difference between changing the page in Inspect vs. a developer actually changing the file on the server?

	Inspect / DevTools	**Actually editing the server file**
What you're editing	A temporary copy in your browser's memory (the "rendered DOM")	The real HTML/CSS/JS file stored on the hosting server
Who sees the change	Only you, only right now	Everyone, everywhere, permanently
Does it survive a refresh?	No — gone instantly	Yes — it's now the new "official" version

Task 7

Comparison Table

Aspect	Swiggy	Zomato
Home Page Layout	Search bar front and center, followed by horizontally scrolling category cards (offers, "Swiggy Instamart," "Dineout"), then restaurant listings in a clean grid below.	Similar structure — big location/search bar at top, followed by cuisine/collection cards, then a restaurant list — but Zomato often surfaces more editorial content (curated collections, "top rated near you") higher up the page.
Colors & Fonts	Bright orange as the primary brand color, white background, rounded sans-serif font — feels energetic and food-forward.	Bright red as the primary brand color, white background, a slightly more condensed/modern sans-serif — feels a touch more premium/editorial.
Search Ease	Search bar is sticky and prominent; typing a dish name (e.g., "biryani") surfaces both restaurants and specific dishes quickly, with filters (rating, price, delivery time) easy to reach.	Search is equally prominent; results are strong for restaurant names but historically leans slightly more toward restaurant-level search than exact-dish search, though this has improved over time.
Responsive/Mobile Behavior	Resizing the window: category cards collapse into a horizontally-swipeable strip, the grid drops from multi-column to single-column, and the search bar stays fixed at the top — feels like a true "mobile-first" layout even on desktop.	Resizes similarly well — cards stack, images scale down, and text reflows — but a few desktop-only elements (like larger banner ads/collections) sometimes feel more "squeezed" rather than truly redesigned for small screens.
Overall Navigation Feel	Very task-focused: gets you from "I'm hungry" to "order placed" with minimal detours.	Slightly more content-rich/browsy: encourages exploring cuisines and curated lists, which is nice for discovery but can add an extra step before ordering.

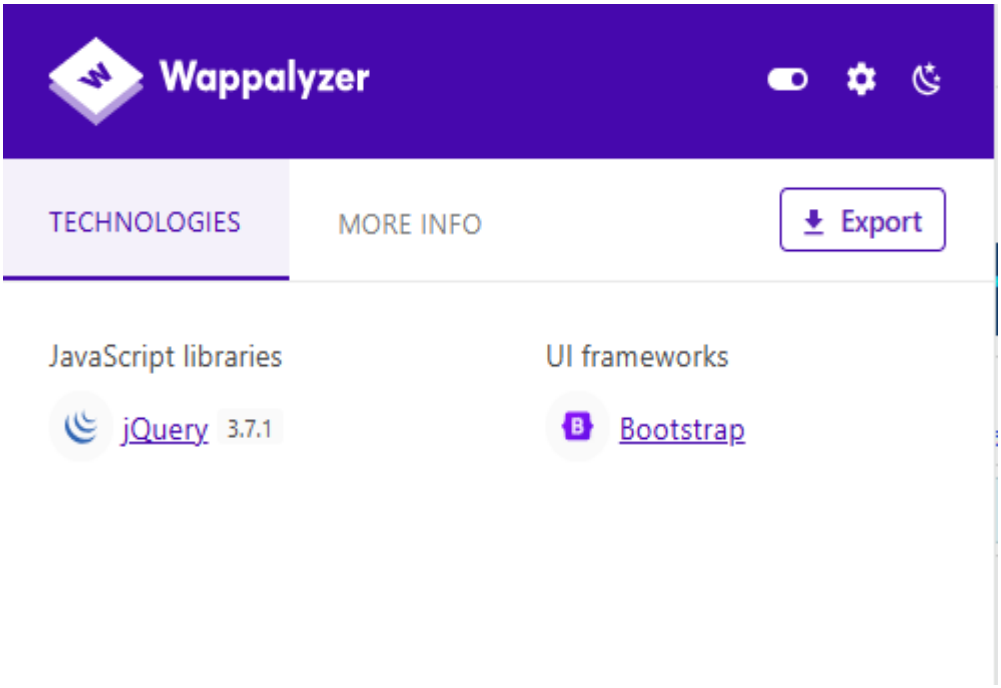
My Personal Opinion

I found Swiggy easier to use — its search-first layout gets me to a specific dish faster, and its mobile resizing felt more deliberately optimized rather than just "shrunk

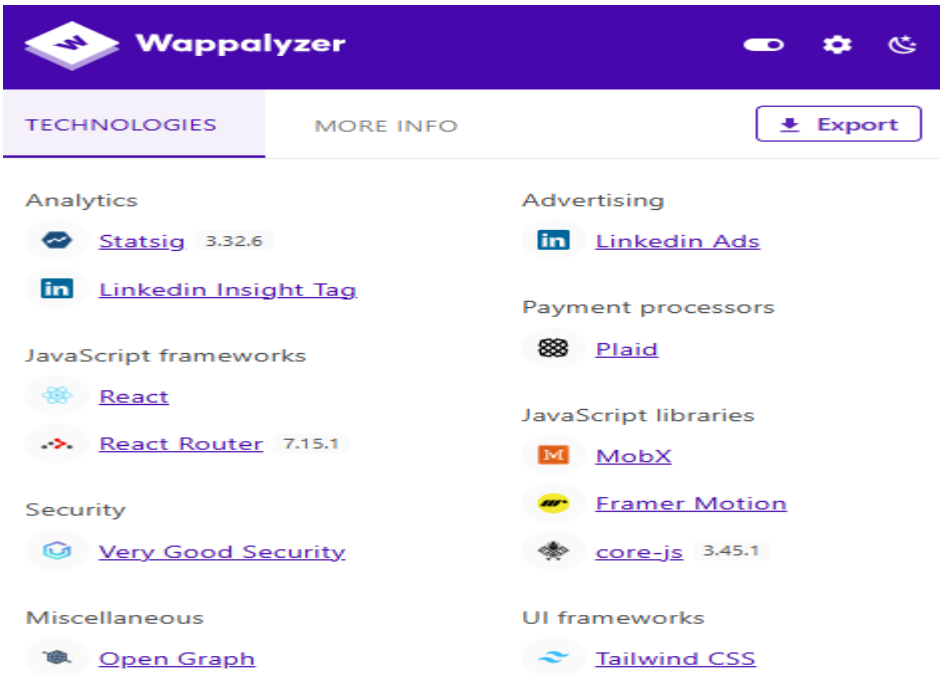
Task 8


Wappalyzer
TECHNOLOGIES
MORE INFO
Export
Analytics
Datadog
99% sure
JavaScript frameworks
React
Astro
Miscellaneous
PWA
Open Graph
HTTP/3
JavaScript libraries
Swiper
Framer Motion
Reverse proxies
Envoy
UI frameworks
Tailwind CSS
RUM
Datadog
99% sure

SBI


Wappalyzer
TECHNOLOGIES
MORE INFO
Export
JavaScript libraries
jQuery 3.7.1
UI frameworks
Bootstrap

Chatgpt



Why might two websites that look similar use different technology behind the scenes?
"Looking similar" only describes the UI layer — the visual design decisions (colors, layout, fonts). What's *underneath* depends on totally different constraints:

Were you surprised by any of the results? Why or why not?
The most striking contrast is between SBI and ChatGPT. A cutting-edge AI product like ChatGPT proudly runs on modern, well-known tools (Next.js, Cloudflare) because speed and developer experience are the priority, and there's little sensitive infrastructure exposed to the frontend anyway. SBI, on the other hand, deliberately reveals almost nothing — no flashy framework name, minimal analytics — because a bank's website is a high-value attack target, and every visible technology is a clue an attacker could exploit.

Task 9

Which site will likely feel fastest, and why: Google will almost certainly win — its homepage is just a logo, a search bar, and a couple of buttons, with barely any images and no ad scripts to load. Fewer files to download = faster "feels loaded" moment.

WiFi vs Mobile Data: WiFi usually wins, but the *gap* tends to be much bigger on the heavy news site than on Google — because Google has so little to download that even a slower connection barely matters, while a news homepage's large images/videos/ad scripts get noticeably throttled on weaker mobile signal.

What slows a page down most: Typically, in order of impact — images and videos (largest file sizes by far), followed by ads and third-party tracking scripts (each one is a separate network request, often from slow external servers), and finally lots of small icons/fonts adding up.

Task 10

Why do almost all websites have a similar footer layout?
It's become a UX convention — once most major sites settled on "logo/tagline → useful links → legal links → copyright" at the bottom, users learned to expect it there. People now instinctively scroll to the footer when they want legal info, contact details, or social links, so deviating from that pattern actually hurts usability rather than showing creativity. It's less about copying each other and more about matching a shared mental model users already have.

Which piece of footer information is legally required?
The Privacy Policy link is the one with real legal teeth — laws like GDPR (EU), CCPA (California), and India's DPDP Act require most websites collecting user data to clearly disclose how that data is used, and a footer link is the standard place to satisfy that requirement. Copyright notices and social links are common practice but not legally mandated the way a privacy policy is.